

ANIL KUMAR TIWARI

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SUMMARY

Data Engineer with 2+ years of experience building scalable data architectures and production-grade ML solutions. Expert in high-performance data engineering pipelines, AI/ML development, and Large Language Model (LLM) integration. Proficient in streamlining processes through autonomous workflow automation with n8n.

TECHNICAL SKILLS

- **Data Engineering:** Apache Airflow, AWS, Azure, Kafka, Spark, PostgreSQL
- **Machine Learning:** TensorFlow, PyTorch, Scikit-learn, YOLOv8, OpenCV, Keras, Pandas, NumPy
- **Cloud & DevOps:** AWS (S3, EC2, Lambda, IAM), Docker, Kubernetes, CI/CD (GitHub Actions), Linux, Terraform
- **Languages:** Python, SQL, JavaScript(React, Next, ReactNative)
- **Concepts:** ETL/ELT Pipelines, Data Warehousing, Object Detection, REST APIs, Agile, n8n, Langchain

EXPERIENCE

Georgia State University | *Graduate Teaching Assistant*

Aug 2025 - Present

- Served as a first point of contact for 50+ students, practicing active listening and providing resource referrals to ensure academic and personal well-being.
- Collaborated with faculty to modernize course curriculum by integrating real-world datasets, ensuring students gained practical experience with industry-standard tools.

Merodream | *Data Engineer*

Sep 2023 - Aug 2025

- Designed and implemented ETL scripts, CTEs, views, procedures, and Redshift tasks to automate the daily data processing orchestrated via Apache Airflow.
- Developed an AWS Redshift pipeline to run daily ETL scripts, monitored batch jobs, and resolved issues, ensuring continuous data processing without downtime
- Optimized SQL queries and implemented table partitioning strategies, cutting cloud compute costs by 25% while handling terabytes of monthly transaction data.
- Collaborated with ML teams to deploy model training pipelines using Docker and Kubernetes, reducing setup time and improving scalability for model training

PROJECTS

SAM2-LatentDiff - SAM2-Guided Low-Light Image Enhancement

- Developed a two-stage low-light enhancement system combining frozen SAM2 hierarchical features with a pretrained Stable Diffusionv1.5 U-Net, using cross-attention and LoRA for parameter-efficient latent-space fine-tuning
- Built an 87K-parameter PixelRefiner to reduce VAE reconstruction artifacts; achieved 18.61 PSNR, 0.7085 SSIM, and 0.2399 LPIPS on the LOL-v2 Real test set.
- Implemented preprocessing, training, evaluation, visualization, and ablation pipelines; demonstrated 111 ms single-pass inference on an NVIDIA A100.

Hull Tactical Market Prediction

- Built a high-performance feature engineering pipeline using Polars and XGBoost (GPU-accelerated), dynamically selecting the top 30 signal features from complex volatility and sentiment interactions.
- Engineered a robust 6-model stacking ensemble (including CatBoost, LightGBM, and Random Forest) with a RidgeCV meta-learner, optimizing model stability for noisy financial data.
- Implemented a production-simulation inference server with volatility targeting logic, ensuring consistent model deployment and risk-adjusted scoring in a live-loop environment.

Wildlife Conservation Image Classification

- Built an automated species classifier using ResNet50 with Transfer Learning to process camera-trap images from Côte d'Ivoire.
- Addressed severe class imbalance (90% empty frames) by implementing Focal Loss and oversampling minority classes, boosting recall for rare species by 20%.
- Developed a "motion-trigger" filtering script to automatically discard empty frames, reducing manual labeling effort by 70%.

EDUCATION

Georgia State University | *M.S., Computer Science*

Aug 2027

Pokhara University | *B.S., Computer Engineering*

Aug 2023

CERTIFICATIONS

- AI Open Source Capstone Course: Codepath (June 2026 - Aug 2026)
- Supervised Machine Learning: Regression and Classification: Stanford/Coursera
- Python for Data Science, AI & Development: IBM (Oct 2023)
- Diploma in Quantum Computing and Programming (Bronze): Qiskit (Sep 2023)
- AWS Academy Graduate – Cloud Foundations: Jul 2023

- Capstone: Retrieving, Processing, and Visualizing Data with Python: Jun 2021
- Python Data Structures: Coursera (Sep 2020)

PUBLICATIONS

- DAAL: Density-Aware Adaptive Line Margin Loss for Multi-Modal Deep Metric Learning.*arXiv:2410.05438*